Subject: PFP191 – Programming Fundamentals with Python

Topic: Book management in the library

Objectives:

Design and implement a Python application to manage student records, applying the concepts learned in the course, including:

- Object-Oriented Programming (OOP).
- Data management using list or dictionary.
- File processing (read/write text).
- Exception Handling.
- Modular Programming (multiple .py files).

Group Work Guidelines

- Group size: 3–5 students.
- Duration: 4–6 weeks.
- Deliverables:
  - Source code (.py).
  - Technical report (REPORT.pdf).
  - Presentation slides (SLIDES.pptx).
- Submission via LMS (as directed by the instructor).

Functional Requirements

1. Book Data Management

- Add / edit / delete a book
- Display book list
- Search for books
- Book information example:

```
class Book:
    __isbn: str
    __title: str
    __author: str
    __category: str
    __publication_year: int
    __is_available: bool = True
```

2. Borrowing and Returning

- Borrow a book
- Return a book
- List of borrowed books

3. Statistics and Reports

- Books by category
- Most borrowed books

4. Console-Based User Interface

- Provide a simple, text-based (CLI) menu system for user interaction.

5. File I/O

- Save/load student data from .txt files.

6. Object-Oriented Programming

- Use classes such as Book and LibraryManager.
- Apply OOP principles: constructors, encapsulation, and inheritance (if needed).

7. Advanced Features

- Organize code into packages: `models/`, `services/`, `utils/`
- Implement `__str__`, and use `@property` for clean object interfaces
- Exception handling for invalid input and file errors.
- Use built-in functions for data manipulation.

Technical Requirements

- Organize the code into multiple .py files (modular design).
- Prepare a presentation explaining the architecture and design decisions.

-- End --.